

Model	Method	Cancer-Myth		CREPE	
		TPR ↑	FPR ↓	TPR ↑	FPR ↓
Qwen2.5 7B	Direct gate	—	—	—	—
	Premise-review CoT gate	—	—	—	—
	GEPA	—	—	—	—
	GEPA + text feedback	—	—	—	—
	GEPA + SAE feedback	—	—	—	—
GPT-6 Luna	Direct gate	—	—	—	—
	Premise-review CoT gate	—	—	—	—
	GEPA	—	—	—	—
	GEPA + text feedback	—	—	—	—
	GEPA + SAE feedback	—	—	—	—
Gemma 4 12B	Direct gate	—	—	—	—
	Premise-review CoT gate	—	—	—	—
	GEPA	—	—	—	—
	GEPA + text feedback	—	—	—	—
	GEPA + SAE feedback	—	—	—	—
Qwen3.8 27B OFF	Direct gate	—	—	—	—
	Premise-review CoT gate	—	—	—	—
	GEPA	—	—	—	—
	GEPA + text feedback	—	—	—	—
	GEPA + SAE feedback	—	—	—	—
Qwen3.8 27B ON	Direct gate	—	—	—	—
	Premise-review CoT gate	—	—	—	—
	GEPA	—	—	—	—
	GEPA + text feedback	—	—	—	—
	GEPA + SAE feedback	—	—	—	—

Table 1. False-premise detection across models and datasets

Unmeasured plan: — is not a measured score. All metrics are percentages.
TPR: false-premise detection rate. FPR: false positives on normal questions.
Feature feedback for detection is evaluated separately from response optimization.